LAB ASSIGNMENT – 1

DATE: 05/01/2026

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RollNo: UG/02/BTCSEAIML/2023/064

RegNO: AU/2023/0010235

Section :'E'

Q1. Create a database named **collegeDB** and switch to it.

```
test> use collegeDB_AIML_064
switched to db collegeDB_AIML_064
```

Q2. Create a collection named **students** in the collegeDB database.

```
collegeDB_AIML_064> db.createCollection("Students_AIML_064")
{ ok: 1 }
```

Q3. Insert **one document** into the students collection with the following fields:

- name
- rollNo
- department
- age

```
collegeDB_AIML_064> db.Students_AIML_064.insertOne({name:"Rahul Sharma",rollNo:101,department:"CSE",age:20})
{
  acknowledged: true,
  insertedId: ObjectId('699ada1220ec1b8c9e628ca0')
}
```

Q4. Insert **four more documents** into the students collection with different values

```
collegeDB_AIML_064> db.Students_AIML_064.insertMany([{name:"Anita Verma",rollNo:102,department:"ECE",age:21},{name:"Vikram Singh",rollNo:103,department:"CSE
",age:22},{name:"Priya Nair",rollNo:104,department:"ME",age:20},{name:"Arjun Patel",rollNo:105,department:"IT",age:23}])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('699ada2320ec1b8c9e628ca1'),
    '1': ObjectId('699ada2320ec1b8c9e628ca2'),
    '2': ObjectId('699ada2320ec1b8c9e628ca3'),
    '3': ObjectId('699ada2320ec1b8c9e628ca4')
  }
}
```

Q5. Display **all documents** present in the students collection.

```
collegeDB_AIML_064> db.Students_AIML_064.find()
[
  {
    _id: ObjectId('699ada1220ec1b8c9e628ca0'),
    name: 'Rahul Sharma',
    rollNo: 101,
    department: 'CSE',
    age: 20
  },
  {
    _id: ObjectId('699ada2320ec1b8c9e628ca1'),
    name: 'Anita Verma',
    rollNo: 102,
    department: 'ECE',
    age: 21
  },
  {
    _id: ObjectId('699ada2320ec1b8c9e628ca2'),
    name: 'Vikram Singh',
    rollNo: 103,
    department: 'CSE',
    age: 22
  },
  {
    _id: ObjectId('699ada2320ec1b8c9e628ca3'),
    name: 'Priya Nair',
    rollNo: 104,
    department: 'ME',
```

Q6. Display only those students who belong to the **CSE department**.

```
collegeDB_AIML_064> db.Students_AIML_064.find({department:"CSE"})
[
  {
    _id: ObjectId('699ada1220ec1b8c9e628ca0'),
    name: 'Rahul Sharma',
    rollNo: 101,
    department: 'CSE',
    age: 20
  },
  {
    _id: ObjectId('699ada2320ec1b8c9e628ca2'),
    name: 'Vikram Singh',
    rollNo: 103,
    department: 'CSE',
    age: 22
  }
]
```

Q7. Update the age of one student (any one) to a new value.

```
collegeDB_AIML_064> db.Students_AIML_064.updateOne({rollNo:101},{$set:{age:25}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

Q8. Delete **one document** from the students collection using any condition

```
collegeDB_AIML_064> db.Students_AIML_064.deleteOne({rollNo:105})
{ acknowledged: true, deletedCount: 1 }
```